Predictive Analytics Using Historical Data

Objective

The objective of this project is to build a predictive analytics model using historical sales data and forecast future sales trends using Machine Learning techniques.

Dataset

The dataset contains yearly sales records from 2015 to 2024.

Columns: Year, Sales

Tools and Technologies Used

- Python
- Pandas
- Matplotlib
- Scikit-Learn

Project Workflow

1. Data Collection – Historical sales data was collected and stored in a CSV file.
2. Data Preprocessing – The dataset was loaded using Pandas and checked for missing values.
3. Model Building – A Linear Regression model was trained using historical sales data.
4. Model Evaluation – Performance was evaluated using MAE and R^2 Score.
5. Future Forecasting – Future sales from 2025 to 2029 were predicted.
6. Data Visualization – Historical sales and predictions were visualized using Matplotlib.

Results

Mean Absolute Error (MAE): 3.75

R^2 Score: 0.9979

Future Sales Predictions:

2025 → 336.38
2026 → 360.95
2027 → 385.52
2028 → 410.09
2029 → 434.66

Conclusion

The Linear Regression model successfully forecasted future sales trends with high accuracy. The project demonstrates predictive analytics, trend analysis, data preprocessing, model evaluation, and forecasting techniques using Python.